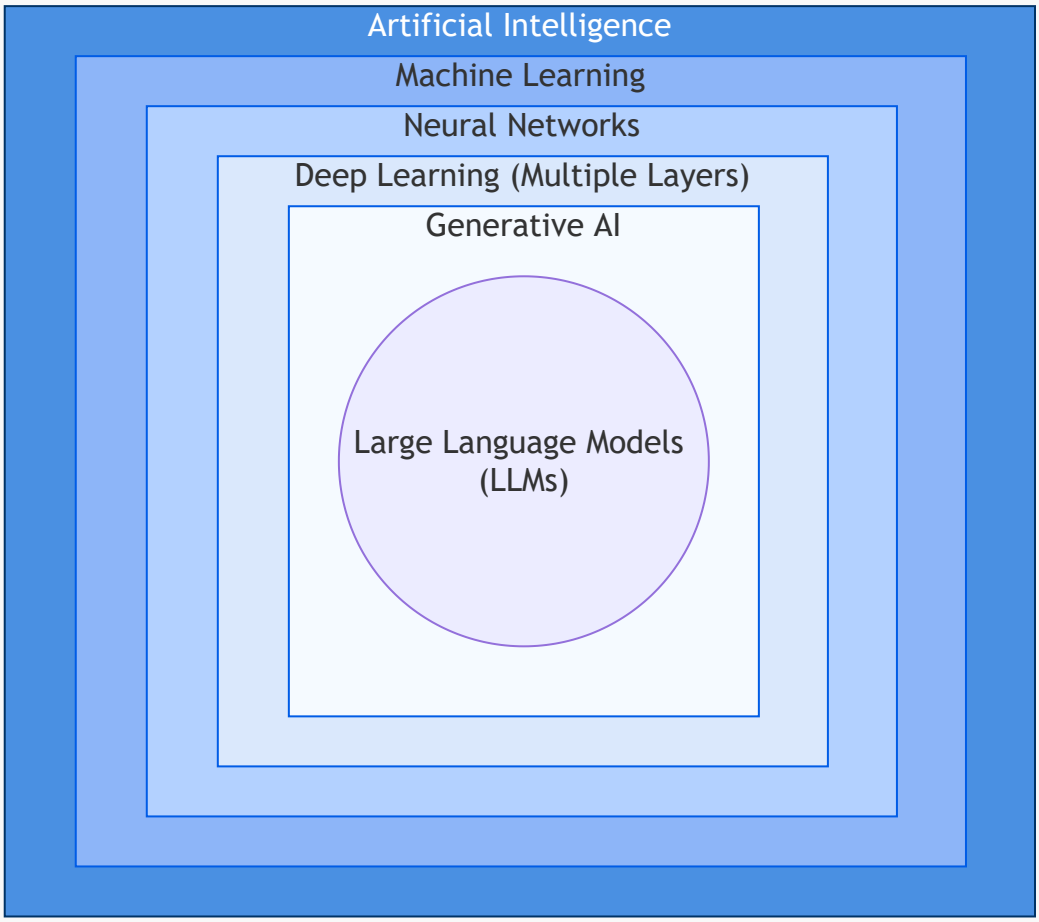
AI outline graph



Definition: Generative AI

讓computer學會產生複雜而有結構的物件, 就是Classification, 因為Generative AI就是給一個未完成的句子, 去猜接下來接哪一個token

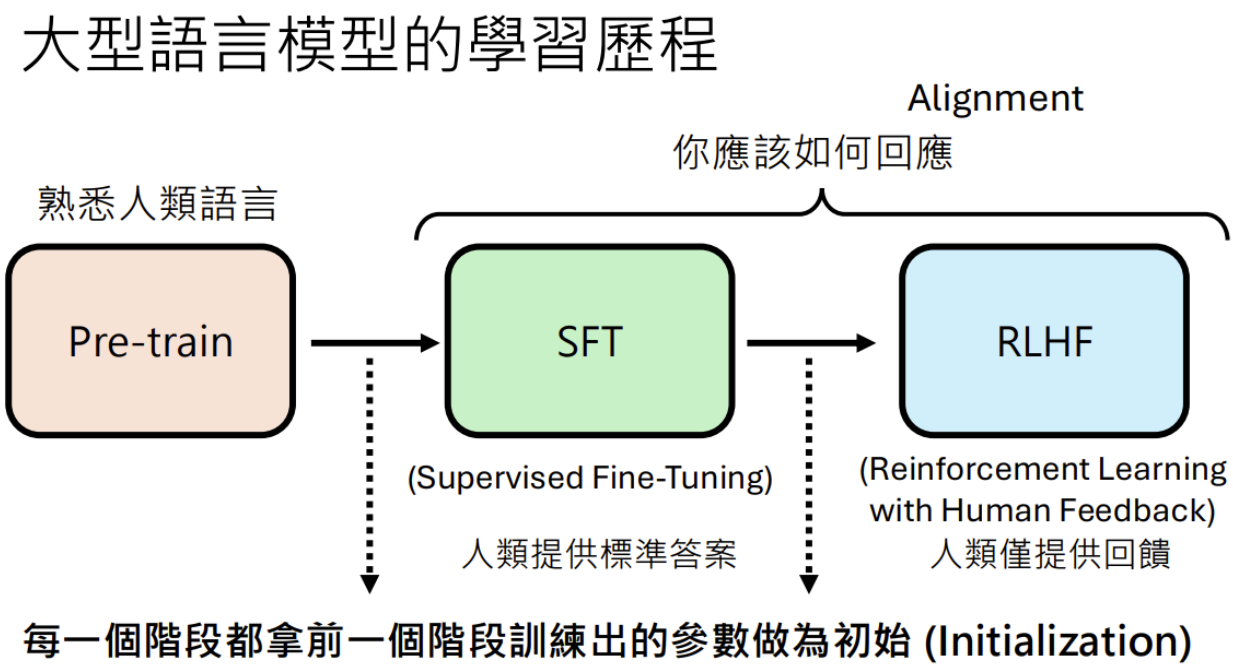
- 複雜 : 無限可能
- 有結構 : 由有限的基本單位(toekn)所構成
- 基本單位(token) :
  - 影像 : 是由像素構成的. 每個像素又由RGB三個子像素組成,而每個子像素的可能值範圍為0-255. 影像只考慮兩個維度: width, height
  - 聲音訊號 : 是由取樣點構成的. 遠看是聲音訊號,拉進後則可以看到一個一個的取樣點.一秒鐘有多少取樣點,取決於取樣率(sampling rate);比如, 16kHz的取樣率意味著一秒鐘有16000個取樣點. 而每個取樣點有多少種可能的數值,則取決於取樣解析度(bit resolution);常見的16-bit解析度有65,536種可能的數值
  - 影片 : 就是一連串的圖片, 每一張圖片又叫Frame. 需考慮三個維度: width, height, time

Retrieval Augmented Generation(RAG)

Context Engineering的核心目標

- 避免塞爆Context(把需要的放進去,不需要的清出來)
- 常用招數
  - Select
    - 挑選需要的內容, e.g. RAG, Tool RAG, Memory RAG
  - Compress
  - Multi-Agent

Machine Learning



0. ML三步驟

設定範圍	考量任務特性，訂出候選函式的集合 Deep Learning (CNN, Transformer ...), Decision Tree, etc.
設定標準	訂出「評量函式好壞」的標準 Supervised Learning, Semi-supervised Learning, RL, etc.
達成目標	找出最好的函式 Gradient Descent (Adam, AdamW ...), Genetic Algorithm, etc.

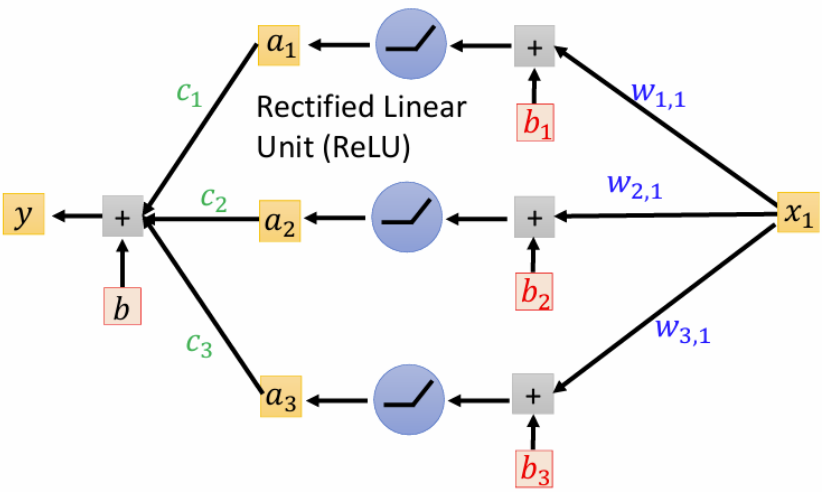
1. prepare data

- split to train data & test data

2. set a model

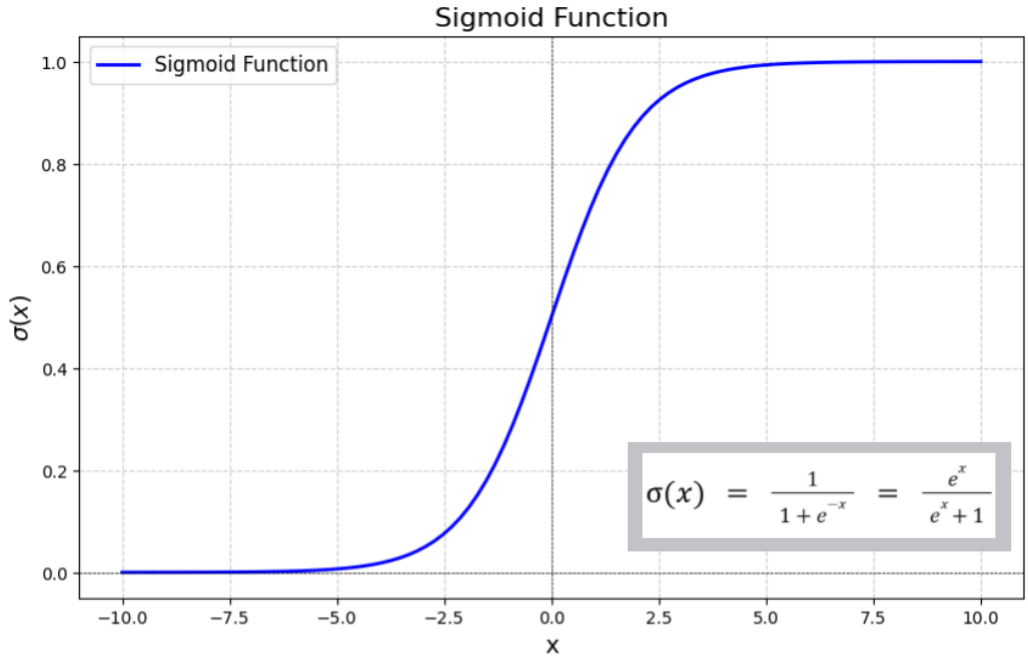
- 根據domain knowledge
  - Simple Linear Regression
    - $y = \beta_0 + \beta_1 x + \epsilon$
      - 其中 β_0 為截距，β_1 為斜率，ϵ 為隨機誤差。
      - y, x is feature.$\beta 1$ is weight.$\beta 0$ is bias
  - Multiple Linear Regression
    - $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n + \epsilon$
  - Piecewise Linear Curves

$$y = b + \sum_{i=1}^{H} c_i \underbrace{max(0, w_{i,1} x_1 + b_i)}_{a_i}$$

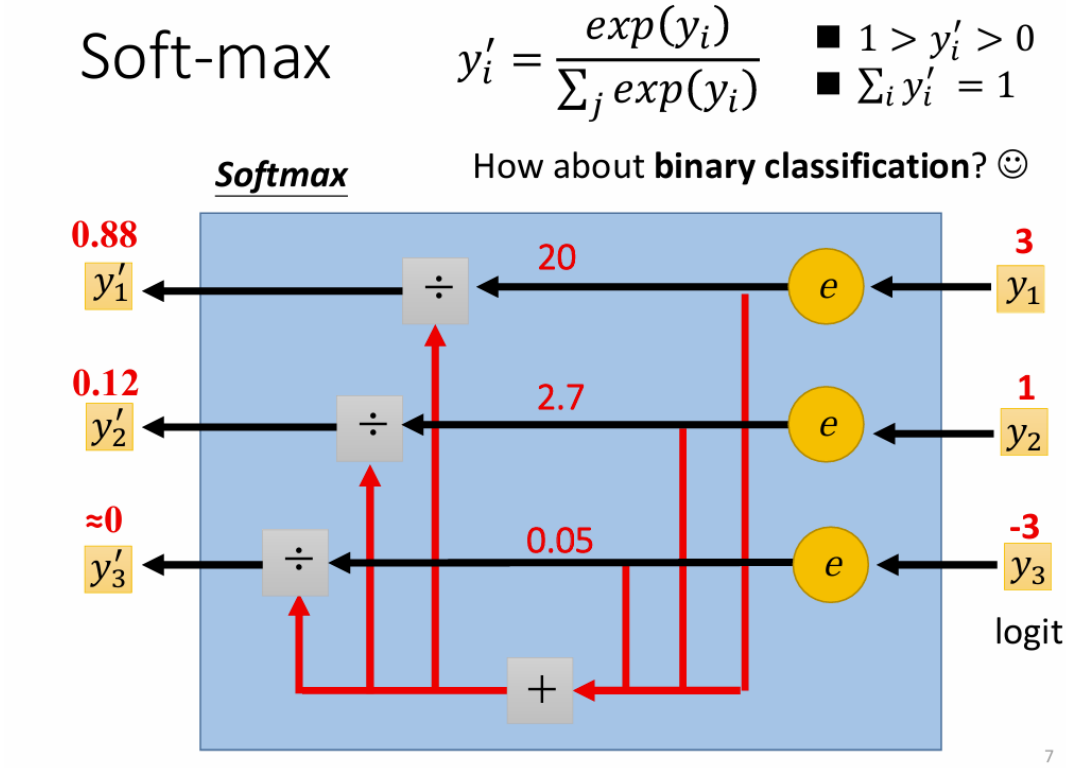


  - Classification

- Binary Classification(Sigmoid=Logistic Function)
  - $P(y=1|x) = \frac{1}{1+e^{-(\beta_0+\beta_1 x_1+\cdots+\beta_n x_n)}}$

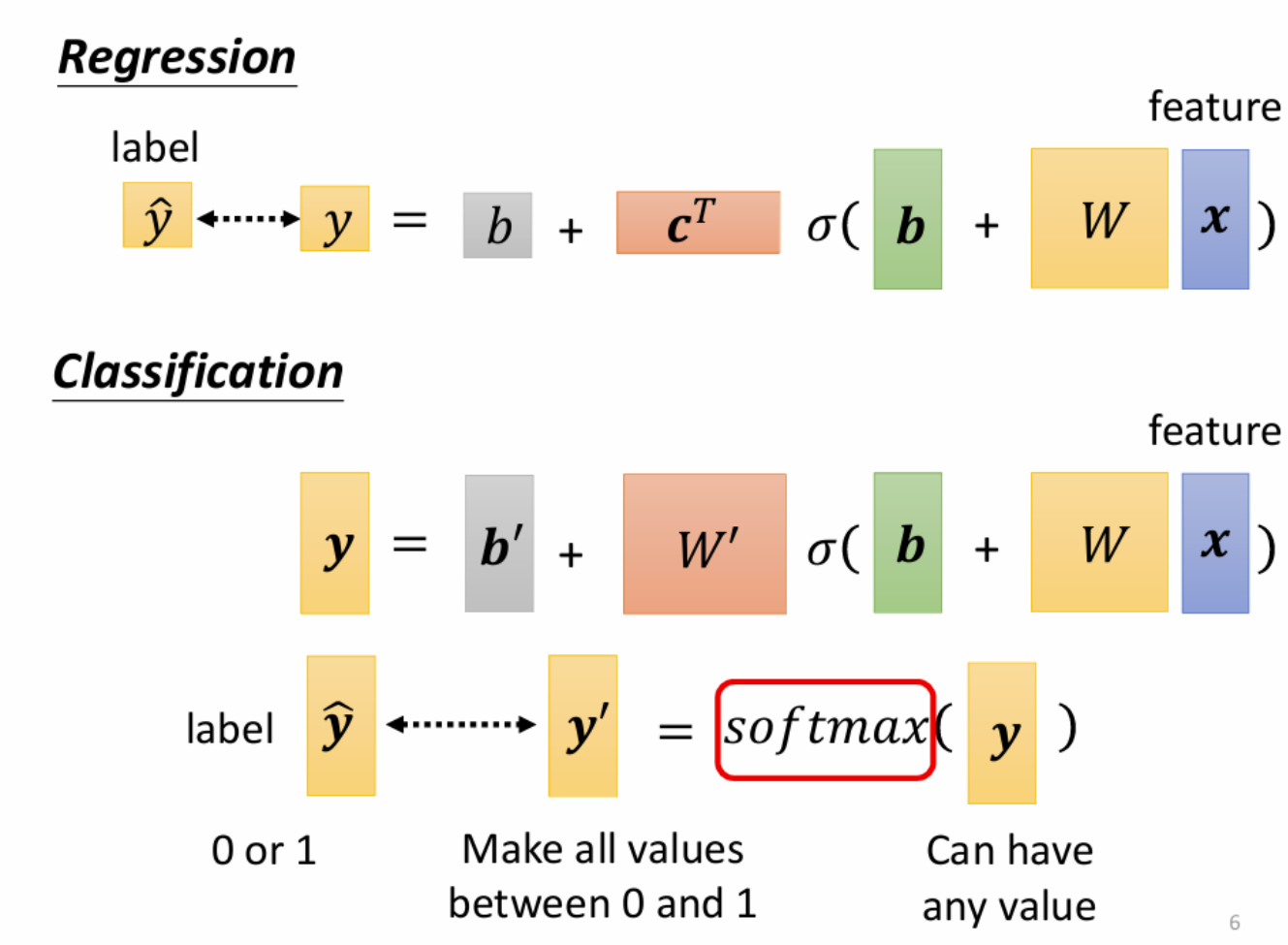


- Multi-class Classification(Softmax)



- Structured Learning
  - **create** something with structure(e.g. image, document)

compare regression and classification



- σ is called **activation function**
  - Regression
    - σ is Relu
  - Classification
    - σ is sigmoid/softmax

3. set Cost function/Loss function

- Mean Squared Error, MSE
  - $MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$
    - n : 訓練資料的總數 (Train Data Size / Batch Size).
    - y_i : 第 i 筆資料的真實標籤 (Ground Truth).
    - $\hat{y}_i$: 模型對第 i 筆資料的預測值 (Prediction)。
- Mean Absolute Error, MAE
  - $MAE = \frac{1}{n}\sum_{i=1}^{n}|(y_i - \hat{y}_i)|$
    - n : 訓練資料的總數 (Train Data Size / Batch Size).
    - y_i : 第 i 筆資料的真實標籤 (Ground Truth).
    - $\hat{y}_i$: 模型對第 i 筆資料的預測值 (Prediction)。
- Cross-Entropy(classification)
  - if $y, \hat{y}$ are both probability distributions
  - $Cross - Entropy = H(y, \hat{y}) = -\sum_{i=1}^{c} y_i \log(\hat{y}_i)$
    - c: The total number of classes or categories.
    - **Minimizing cross-entropy** is equivalent to **maximizing likelihood**

- y_i: The probability of the i-th class in the true distribution (often a one-hot encoded vector where only one $y_i = 1$ and others are 0).
- $\hat{y}_i$: The probability of the i-th class predicted by the model (usually the output of a Softmax function($0 \le \hat{y}_i \le 1$ and $\sum \hat{y}_i = 1$.)).

4. set optimizer

- Gradient Descent(Vanilla Gradient Descent)
  - $\theta^{t+1} = \theta^t - \eta \cdot \nabla L(\theta^t)$
    - $\nabla L(w)$ 代表損失函數 L 在當前參數 w 位置的「斜率」或「坡度」
    - η : The Learning Rate
  - 問題 : 只根據當下算出來的g^t來決定方向
    - 解法 : Gradient Descent + Optimizer
      - 根據$g^0, g^1, g^2, \ldots, g^t$ 一起來決定方向(調整learning rate)
        - Adagrad
          - for each dimension i : $\eta = \frac{\eta}{\sigma_i^t}$, $\sigma_i^t = \sqrt{\sum_{i=0}^{t} (g_i^t)^2}$
          - 問題 : 每一個維度權重相同
        - RMSProp
          - 解決Adagrad : 最近算出來的gradient給比較大的影響
          - use α
  - 可以用 Momentum (下滑之物體會有動量繼續前行, 不會直接殺停)來繼續找 global minimum用以脫離saddle point or local minimum
    - for each dimension i : $\theta^{t+1} = \theta^t - \eta \cdot m_i^t$, $m_i^t = g_i^0 + g_i^1 + g_i^2 + \ldots + g^t$
  - Adam : RMSProp + Momentum

➢ Compute gradient $\boldsymbol{g}^t = \nabla L(\boldsymbol{\theta}^t)$

Momentum

For each dimension i: $m_i^t = \beta m_i^{t-1} + (1-\beta) g_i^t$ $\qquad$ $\theta_i^{t+1} \leftarrow \theta_i^t - \eta m_i^t$

RMSProp

For each dimension i: $\sigma_i^t = \sqrt{\alpha \left(\sigma_i^{t-1}\right)^2 + (1-\alpha)\left(g_i^t\right)^2}$ $\qquad$ $\theta_i^{t+1} \leftarrow \theta_i^t - \frac{\eta}{\sigma_i^t} g_i^t$

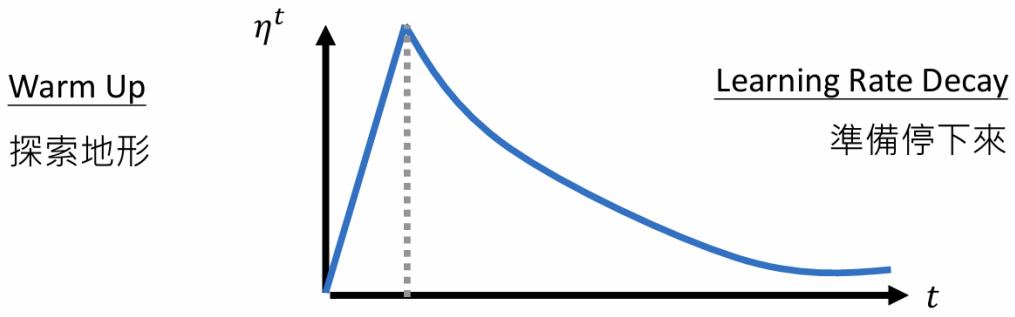
Adam

$$\theta_i^{t+1} \leftarrow \theta_i^t - \frac{\eta}{\sigma_i^t} m_i^t$$

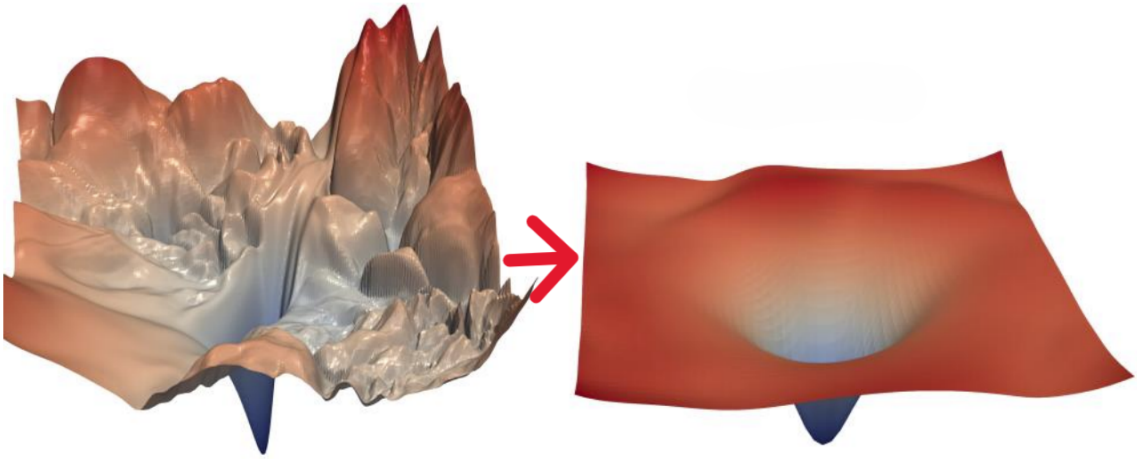
Adam has bias-corrected terms, which are omitted here for simplicity.

- 也可以用Learning Rate Scheduling來調整learning rate

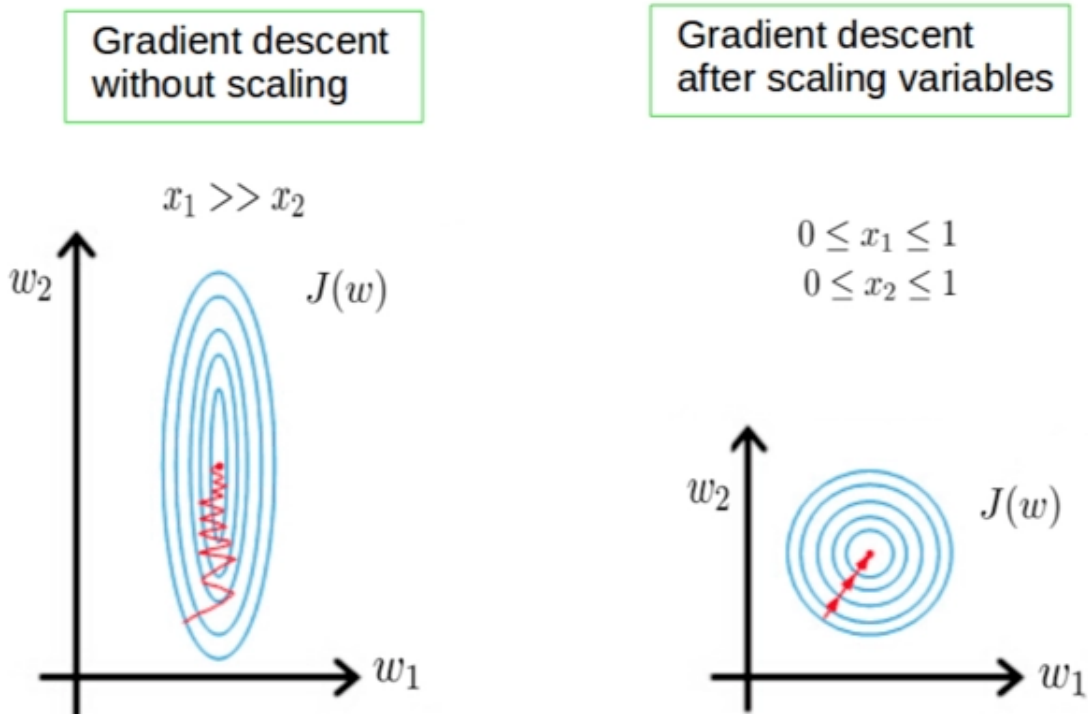
$$\theta_i^{t+1} \leftarrow \theta_i^t - \frac{\eta}{\sigma_i^t} m_i^t \quad \Rightarrow \quad \theta_i^{t+1} \leftarrow \theta_i^t - \frac{\boxed{\eta^t}}{\sigma_i^t} m_i^t$$



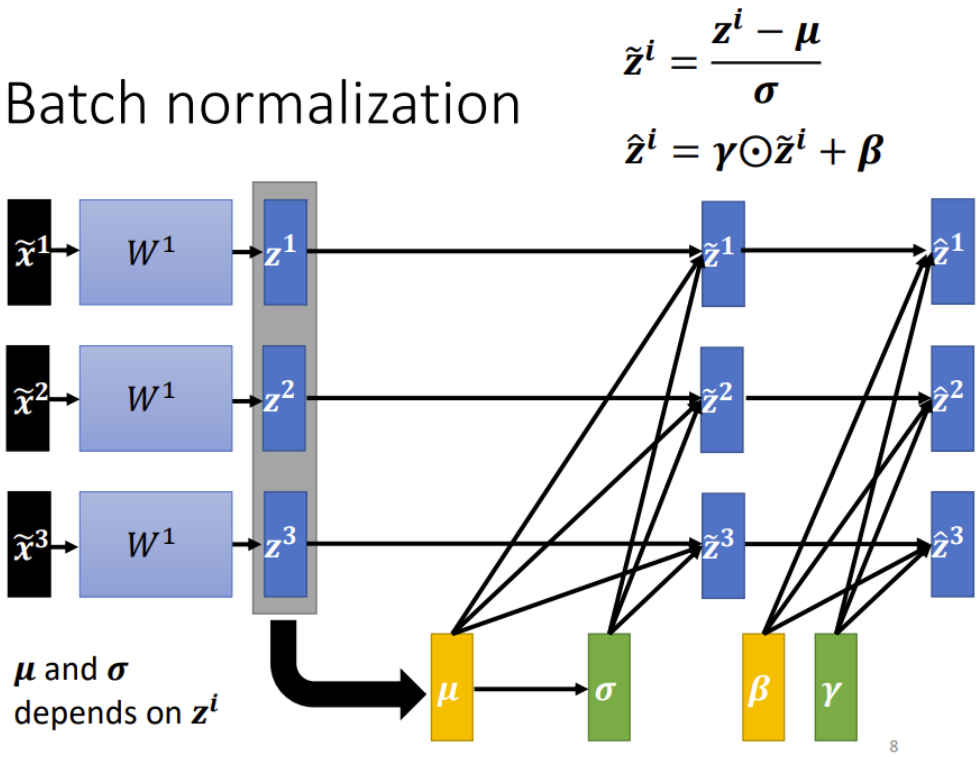
  - warp up : 給optimizer探索地形的機會, 因為剛進入一個新地圖, 不知道地圖有什麼, 設定一個大的learning rate, 讓參數亂跑, 可以大概知道地圖長什麼樣
- Feature Scaling(夷平error surface)



  - accelerate gradient descent



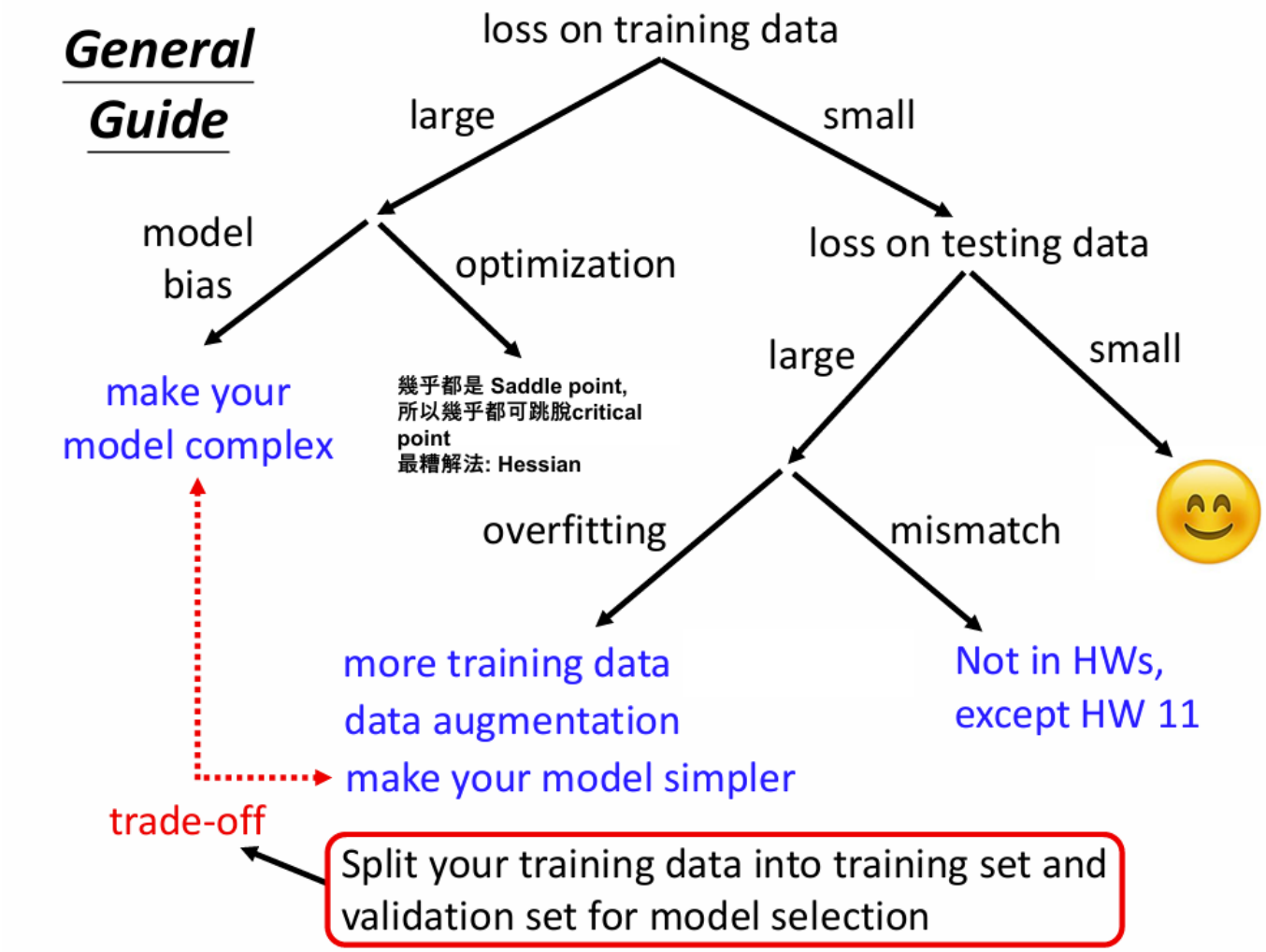
- Normalization
  - Batch Normalization



    - γ, β are another network parameters, 另外再被learned出來的;因為normalization完後,$\tilde{z}^1, \tilde{z}^2, \tilde{z}^3, \ldots$ 之平均為0, 可能會對模型產生限制,所以加上 γ, β
    - γ initialize to $[1, 1, \ldots, 1]^T$
    - β initialize to $[0, 0, \ldots, 0]^T$
  - Layer Normalization
- Standardizatoin

5.Train the model

- Initialization
  - Kaiming Initialization
- **Use gradient descent to train the model, accelerating convergence to the minimum loss and yielding the optimal model(Find the best $\beta_0, \beta_1, \beta_2, \beta_3, \ldots, \epsilon$).**
- general guide



6. use validation data to evaluate your model

- be aware of overfitting

7. use test data to test/inference your model

- public test data
  - be aware of overfitting
- private test data